



Changing the subject with more complex formula

Task A: Changing the subject with two steps

1) Rearrange each formula to make c the subject.

a) $6c + 7 = 19$

$$6c = 19 - 7$$

$$c = \frac{19 - 7}{6}$$

b) $3c + 8 = d$

$$3c = d - 8$$

$$c = \frac{d - 8}{3}$$

c) $3d - c = 19$

$$3d = 19 + c$$

$$3d - 19 = c$$

d) $d - 3c = 19$

$$d = 19 + 3c$$

$$d - 19 = 3c$$

$$\frac{d - 19}{3} = c$$

e) $\frac{c}{9} + 3 = d$

$$\frac{c}{9} = d - 3$$

$$c = 9(d - 3) = 9d - 27$$

f) $\frac{c + 3}{9} = d$

$$c + 3 = 9d$$

$$c = 9d - 3$$

2) Laura has been rearranging these equations to make y the subject.

In each case circle Laura's errors and write a sentence to explain what she should do differently.

a) $2y + 3x = 18$

$$2y = 18 - 3x$$

For y to be the subject the coefficient needs to be 1. Divide both sides by 2

b) $\frac{y}{2} - 5 = x$

$$y - 5 = 2x$$

$$y = 2x + 5$$

When multiplying both sides by 2 be sure to multiply all terms. This line should read $y - 10 = 2x$

c) $2y + 3 = 7x$

$$2y = 7x + 3$$

$$y = \frac{7x + 3}{2}$$

You meant to add -3 to both sides



I have checked my answers and they are all wrong. Can you help me spot where I went wrong?



Task B: Rearranging more complex formula

 1) Rearrange each formula to make x the subject.

$$\begin{aligned} \text{a)} \quad y &= 4x^2 \\ \frac{y}{4} &= x^2 \\ \sqrt{\frac{y}{4}} &= x \end{aligned}$$

$$\begin{aligned} \text{b)} \quad y &= (4x)^2 \\ \sqrt{y} &= 4x \\ \frac{\sqrt{y}}{4} &= x \end{aligned}$$

$$\begin{aligned} \text{c)} \quad y &= 4x^2 + 1 \\ y - 1 &= 4x^2 \\ \frac{y - 1}{4} &= x^2 \\ \sqrt{\frac{y - 1}{4}} &= x \end{aligned}$$

$$\begin{aligned} \text{d)} \quad y &= 4(x^2 + 1) \\ \frac{y}{4} &= x^2 + 1 \\ \frac{y}{4} - 1 &= x^2 \\ \sqrt{\frac{y}{4} - 1} &= x \end{aligned}$$

$$\begin{aligned} \text{e)} \quad y &= \frac{x^2}{4} \\ 4y &= x^2 \\ \sqrt{4y} &= x \end{aligned}$$

$$\begin{aligned} \text{f)} \quad y &= \frac{4}{x^2} \\ x^2 y &= 4 \\ x^2 &= \frac{4}{y} \\ x &= \sqrt{\frac{4}{y}} \end{aligned}$$

$$\begin{aligned} \text{g)} \quad 3x &= \sqrt{y} \\ x &= \frac{\sqrt{y}}{3} \end{aligned}$$

$$\begin{aligned} \text{h)} \quad \sqrt{3x} &= y \\ 3x &= y^2 \\ x &= \frac{y^2}{3} \end{aligned}$$

$$\begin{aligned} \text{i)} \quad 3\sqrt{x} &= y \\ \sqrt{x} &= \frac{y}{3} \\ x &= \left(\frac{y}{3}\right)^2 \\ x &= \frac{y^2}{9} \end{aligned}$$

